



TM5 Review Guide for Test 5 on 04.16

Completion requirements

Automated Model Selection and Occam's Razor

- Terms to know:
 - Pipeline (conceptual pipeline, not just the sklearn Pipeline class!)
 - Hyperparameter tuning
 - Grid search
 - K-fold cross validation
- You should understand how k-fold cross-validation works (we covered this earlier in the course before revisiting it—you may need to look back at your earlier notes and reading for details).
- You should understand what is meant by grid search and how the process works. In particular, you should understand the tradeoffs as far as time.

Support Vector Machines (SVM)

- Terms to know:
 - **Support Vector Machine (SVM)**
 - **hyperplane**
 - **kernel**
 - **margin**
 - **support vectors**
 - **soft margin**
 - **hard margin**
 - **adaptive vs. fixed basis function**
 - **lifting**
 - **C hyperparameter**
 - **gamma hyperparameter**
- SVM is a **supervised learning** algorithm that can be used for both classification and regression problems.
- SVM tries to find the best hyperplane that separates the data into different classes.
- The best hyperplane is the one that maximizes the margin between the two classes.
- The margin is the distance between the hyperplane and the closest data points from each class.
- The data points that are closest to the hyperplane are called **support vectors**.

- SVM can handle both linearly separable and non-linearly separable data by using a **kernel** function to transform the data into a higher-dimensional space where it is linearly separable.
- The process of transforming the data into a higher-dimensional space is also known as **lifting**. Lifting involves mapping the original data into a higher-dimensional space using a set of non-linear functions.
- The kernel function is responsible for performing the lifting, allowing the SVM algorithm to find a hyperplane that separates the data in this higher-dimensional space.
- The most commonly used kernel functions are:
 - **linear kernel**
 - **polynomial kernel**
 - **radial basis function (RBF) kernel**
- SVM can also handle cases where the data is not perfectly separable by using a **soft margin**.
- The soft margin allows some data points to be misclassified in order to find a better hyperplane that separates the majority of the data correctly.
- The degree of misclassification allowed is controlled by the **C hyperparameter**. A smaller C value allows more misclassification, while a larger C value allows less misclassification.
- The **gamma hyperparameter** controls the shape of the kernel function. A smaller gamma value makes the kernel function wider, while a larger gamma value makes the kernel function narrower.
- Lifting can be computationally expensive, but it can be a powerful technique for handling complex data that is not linearly separable in the original feature space.
- SVM is a powerful algorithm that can be used for a wide range of problems, but it can also be computationally expensive and sensitive to the choice of parameters.

Clustering and K-Means

- Terms to know:
 - **Supervised** vs. **unsupervised** machine learning
 - **clustering**
 - **cluster**
 - **centroid**
 - seeds (in the context of choosing the initial points for the K means)
- We have already studied several **distance metrics** in the course. This is a good opportunity to review them.
- Given several data points, each with 3-4 features, you should be able to find the centroid (mean of multivalued variables).

- Example: What is the centroid of the points (2, 0, 8), (10, 4, 0), and (0, 2, 1)?
Answer: (4, 2, 3).
- You should know how the kMeans algorithm works and be able to trace the steps as it clusters a very simple set of data.
- You should understand the difference between a **local** minimum (or maximum) and a **global** maximum and that algorithms such as kMeans can get "stuck" in a local minimum without converging to the global minimum. (See the example in the slides and make sure you understand it.)
- You should be prepared to compare and contrast kMeans with kNN. (In particular, be careful that you do not mix them up, since that is a common mistake on the final exam!)

Dimensionality Reduction and PCA

- Terms to know:
 - dimensionality reduction
 - oblique projection
 - mean
 - variance
 - orthogonal
 - principal component analysis (PCA)
- You should also be familiar with the terms **eigenvector** and **eigenvalue** in the context of PCA. However, I am not expecting a rigorous mathematical definition.
- Obviously, we are not going to ask you to compute the eigenvalues and eigenvectors on a test!
- You should have a general grasp of what PCA does:
 - It transforms your data into a new coordinate system, usually with fewer axes.
 - The new axes are orthogonal (at right angles) to each other everywhere.
 - The new axes are in a particular order: the first axis (principal component) is the one that maximizes the variance in the data. Subsequent components are chosen in the same way, with the constraint that they must be at right angles to the previously selected components.
 - The new axes are linear combinations of all of the original axes (features or dimensions).
- You should know that PCA is a form of **unsupervised learning** and that it is often used as a preprocessing step in preparation for classification or regression learning.
- You should be prepared to relate PCA to:
 - other forms of unsupervised learning (in particular K means);
 - other forms of data preparation (such as scaling, discretization, and one-hot encoding);

- other forms of data projection, such as logistic regression and random projections that are sometimes used in random forests.

Time Series and ARIMA models

- Terms to know:
 - Discrete time
 - forecasting
 - Box-Jenkins methodology
 - (long-term) trend
 - seasonal (pattern)
 - cyclic (pattern)
 - ARIMA (know what it stands for)
 - stationary process
 - Markov assumption
 - White noise process
 - Autoregressive model of order p [AR(p)]
 - Moving average model of order q [AR(q)]
 - ARMA model, ARIMA, VARIMA, etc.
 - differencing
 - extrapolation
 - causal model
- You should understand that time series data require special techniques and objectives.
- You should understand that while we may employ train-test split with a time series, we don't randomize it in the same way as with general supervised machine learning problem since we have to take sequence into account.
- Be sure you know the statistical assumptions of the model
- Earlier in the course we talked about independence assumptions. You should have a general understanding of how time series models exploit these assumptions. For example:
 - Markov assumption: We assume that the variable under observation y_t depends on its value only over a small number of recent time steps $t-1, t-2, \dots, t-k$.
- Make sure you are familiar with Box-Jenkins models in general and ARIMA models in particular. You should know how to fit and evaluate the model.
- Finally, make sure you understand why we call these **autoregressive** models: it is regression in which the variable under observation depends on itself. This is common with time series models.

Final note

- For all ML methods, you should be prepared to discuss the tradeoffs between the various methods and to identify the parameters and hyperparameters of the method.

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